



National Forensic Sciences University, Goa Campus
M.Tech. 1st Sem
Term Assessment-2

Final Date of Submission: 25th Nov 2025
Hard Copy and Handwritten Copy will be considered only

Subject Code: CTMTAIDS S1 P2 Subject Name: Network Security/Network Security and Forensics Instructions - 1) Answer all questions. 2) Assume suitable data. 3) Scientific Calculator is allowed. 4) Parts of the question should attend the same place. Total Marks: 25		
Q.1	Q.1 Short Answer Questions (2 Marks Each × 5 = 10 Marks) Attempt all questions.	Marks
	(a) What is the role of “Order of Volatility” in digital forensics? Explain with an example. (b) Define <i>Man-in-the-Middle (MITM) Attack</i> . Mention two common techniques used to perform it. (c) What is a Security Baseline? How is it useful in securing enterprise systems? (d) Explain the purpose of a DMZ (Demilitarized Zone) in network architecture. (e) What is SQL Injection? Briefly describe how parameterized queries help prevent it.	10
Q.2	Scenario-Based and Practical Questions (2 Marks Each × 5 = 10 Marks)	

	Attempt all questions. (a) During a wireless security assessment, you detect multiple rogue access points. • List three methods to validate and classify these rogue APs. (b) You are assigned to conduct forensic analysis on a compromised Linux server. • Which commands and log files would you check first? • Justify your answer briefly. (c) A system admin is tasked with setting up Wireshark for real-time monitoring. • Mention three filter expressions and explain their usage. (d) As part of an OS hardening process, list three best practices to secure a Windows Server against remote exploits. (e) A network engineer wants to verify SSL/TLS communication on port 443. • Provide three <i>tcpdump</i> commands/filters suitable for analyzing HTTPS traffic.	10
Q.3	Long Answer / Analytical Question (5 Marks) Attempt All	5
	a) As a security analyst at NFSU Goa, you are asked to set up a Security Operations Center (SOC). Write a detailed note on: • Core functions of SOC • Integration with SIEM tools • How SOC aids in real-time incident response Include one real-life use case in your answer. (b) You are asked to implement a Zero Trust Security Model in a mid-sized enterprise. Write a detailed note on: • Key principles of Zero Trust • Role of Network Segmentation and Identity Management • Integration with continuous monitoring tools Include one real-world incident where Zero Trust could have prevented an attack.	

~~~~~END OF PAPER~~~~~